

Overview of Medical image computing

Subject: Medical image computing

1.

Shape correspondence is the methodology that computes correspondent locations between geometric shapes represented by triangle meshes, contours, point sets or volumetric images. Obviously definition of correspondence will influence directly the analysis. Among the different options for correspondence frameworks are: anatomical correspondence, manual landmarks, functional correspondence (i.e. in brain morphometry locus responsible for same neuronal functionality), geometry correspondence, (for image volumes) intensity similarity, etc. Some approaches, e.g. spectral shape analysis, do not require correspondence but compare shape descriptors directly. Statistical analysis will provide measurements of structural change at correspondent locations.

2.

In longitudinal image processing, segmentation and analysis methods of individual time points are informed and regularized with common information usually from a within-subject template. This regularization is designed to reduce measurement noise and thus helps increase sensitivity and statistical power. At the same time over-regularization needs to be avoided, so that effect sizes remain stable. Intense regularization, for example, can lead to excellent test-retest reliability, but limits the ability to detect any true changes and differences across groups. Often a trade-off needs to be aimed for, that optimizes noise reduction at the cost of limited effect size loss. Another common challenge in longitudinal image processing is the, often unintentional, introduction of processing bias. When, for example, follow-up images get registered and resampled to the baseline image, interpolation artifacts get introduced to only the follow-up images and not the baseline. These artifact can cause spurious effects (usually a bias towards overestimating longitudinal change and thus underestimating required sample size). It is therefore essential that all-time points get treated exactly the same to avoid any processing bias. Post-processing and statistical analysis of longitudinal data usually requires dedicated statistical tools such as repeated measure ANOVA or the more powerful linear mixed effects models. Additionally, it is advantageous to consider the spatial distribution of the signal. For example, cortical thickness measurements will show a correlation within-subject across time and also within a neighborhood on the cortical surface - a fact that can be used to increase statistical power. Furthermore, time-to-event (aka survival) analysis is frequently employed to analyze longitudinal data and determine significant predictors.

3.

Atlases Medical images can vary significantly across individuals due to people having organs of different shapes and sizes. Therefore, representing medical images to account for this variability is crucial. A popular approach to represent medical images is through the use of one or more atlases. Here, an atlas refers to a specific model for a population of images with parameters that are learned from a training dataset. The simplest example of an atlas is a mean intensity image, commonly referred to as a template. However, an atlas can also include richer information, such as local image statistics and the probability that a particular spatial location has a certain label. New medical images, which are not used during training, can be mapped to an atlas, which has been tailored to the specific application,

such as segmentation and group analysis. Mapping an image to an atlas usually involves registering the image and the atlas. This deformation can be used to address variability in medical images.

4.

Additional notes Medical image computing is also related to the field of computer vision. An international society, The MICCAI Society, represents the field and organizes an annual conference and associated workshops. Proceedings for this conference are published by Springer in the Lecture Notes in Computer Science series. In 2000, N. Ayache and J. Duncan reviewed the state of the field.

5.

Segmentation is the process of partitioning an image into different meaningful segments. In medical imaging, these segments often correspond to different tissue classes, organs, pathologies, or other biologically relevant structures. Medical image segmentation is made difficult by low contrast, noise, and other imaging ambiguities. Although there are many computer vision techniques for image segmentation, some have been adapted specifically for medical image computing. Below is a sampling of techniques within this field; the implementation relies on the expertise that clinicians can provide.

6.

Image registration is a process that searches for the correct alignment of images. In the simplest case, two images are aligned. Typically, one image is treated as the target image and the other is treated as a source image; the source image is transformed to match the target image. The optimization procedure updates the transformation of the source image based on a similarity value that evaluates the current quality of the alignment. This iterative procedure is repeated until a (local) optimum is found. An example is the registration of CT and PET images to combine structural and metabolic information (see figure). Image registration is used in a variety of medical applications:

7.

Task related fMRI is acquired as the subject is performing a sequence of timed experimental conditions. In block-design experiments, the conditions are present for short periods of time (e.g., 10 seconds) and are alternated with periods of rest. Event-related experiments rely on a random sequence of stimuli and use a single time point to denote each condition. The standard approach to analyze task related fMRI is the general linear model (GLM). Resting state fMRI is acquired in the absence of any experimental task. Typically, the objective is to study the intrinsic network structure of the brain. Observations made during rest have also been linked to specific cognitive processes such as encoding or reflection. Most studies of resting state fMRI focus on low frequency fluctuations of the fMRI signal (LF-BOLD). Seminal discoveries include the default network, a comprehensive cortical parcellation, and the linking of network characteristics to behavioral parameters. There is a rich set of methodology used to analyze functional neuroimaging data, and there is often no consensus regarding the best method. Instead, researchers approach each problem independently and select a suitable model/algorithm. In this context there is a relatively active exchange among neuroscience, computational biology, statistics, and machine learning communities. Prominent approaches include

8.

Multiple templates Rather than relying on a single template, multiple templates can be used. The idea is to represent an image as a deformed version of one of the templates. For example, there could be one template for a healthy population and one template for a diseased population. However, in many applications, it is not clear how many templates are needed. A simple albeit computationally expensive way to deal with this is to have every image in a training dataset be a template image and thus every new image encountered is compared against every image in the training dataset. A more recent approach automatically finds the number of templates needed.